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Sustainability-4-HRI, HRI-4-Sustainability

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***Abstract*—"Sustainability-4-HRI, HRI-4-Sustainability" offers hands-on engagement with the HRI 2025 conference theme, "Robots for a Sustainable World". This workshop will explore the relationship between HRI and sustainable development and stimulate discussion on how we can make our own research practices more sustainable. We propose a full-day workshop, featuring morning discussions with sustainability experts and activists, and an afternoon hands-on activity aimed at understanding how robotics research can help in creating sustainable futures. We will broadly answer the following questions: How can we, as robotic researchers, help address sustainable development responsibly? Equally, how can we ensure that our HRI research practices minimises ecological footprints and operates within ethical and sustainable frameworks? We envision two practical outcomes of this workshop: a "sustainability statement" that can be submitted together with future HRI research papers, and a paper gathering insights and reflections from the workshop. We welcome researchers and students at any career stage and from any subfield of HRI to attend and contribute.**

I. Introduction

This workshop aims to engage the HRI community in a critical examination of sustainability within our field. Our approach extends beyond the narrow view of "robots for environmental applications" to consider the multifaceted impact of HRI research and its potential to foster a more sustainable world — socially, ecologically, and technologically. We define sustainability in HRI as encompassing two key dimensions:

- Sustainable research practices: resource management, open science, transparency, and ethical considerations throughout the robot design, development, and deployment pipeline.
- HRI's contribution to global sustainability efforts: how can our field address the UN Sustainable Development Goals (SDGs) and broader societal challenges?

Our exploration of these questions is fundamentally motivated by a desire (and, given the state of the world, a pressing need) to engage the HRI community in thinking more broadly about the impact of robot design, development and deployment. We propose that sustainability in HRI is multifaceted and deeply interconnected with real-world needs and ethical responsibilities. Through this perspective, sustainability complements recent movements around e.g. Feminist HRI [1], robots for social justice [2] and critical robotics [3], [4] in addition to broader topics of e.g. the social shaping and politics of technology [5] and trustworthy and socially sustainable artificial intelligence.

The workshop is designed to achieve three interconnected objectives:

1) **Broadening the Concept of Sustainability in HRI**: we aim to encourage researchers to think beyond conventional applications and consider the wider impact of our work on society and the planet.
2) **Critical Mapping of HRI Components**: participants will examine the entire HRI pipeline through a sustainability lens, reflecting on research practices, resource management, and ethical considerations.
3) **Interdisciplinary Dialogue**: we seek to connect HRI back to the "real world" by inviting diverse perspectives, including from non-technical fields and from activists, to address pressing global needs.

We want to engage in exploration of these topics with *care*. Climate anxiety is real, and we want to provide a space for researchers, especially early-career scholars, to discuss and navigate the tensions between technological innovation and sustainability concerns, to wrestle with any conflict they might experience between wanting to make the world a better place and the reality of developing resource-intensive technologies —"feature-heavy electronic devices meeting the imaginary needs of imaginary users" [6]. As researchers we can support each other to critically analyze robotic research as a phenomenon, and explore research methods and approaches that support a sound critical discourse on the role and emergence of robotics in society [3], [4] to address our responsibility in the politics of technology and shaping of society [5].

Through collaborative activities, we plan to develop a conceptual framework for sustainability in HRI, addressing both the research process and its intended outcomes. We hope the workshop can be a first step towards developing a community as a site of resilience, change, reflection, and mutual support.

II. Tentative program

We propose a full day workshop, featuring knowledge exchange (presentations from invited speakers and panel discussion) and hands-on activities aimed at generating specific outcomes (more details on these below). Tentative schedule:

- 9:00-9:30. Meet and greet, introduction from organisers
- 9:30-10:30. Flash talks from invited speakers
- 10:30-11:30. Panel and Q&A with speakers
- 11:30-12:00. Coffee break
- 12:30-13:00. SDGs and Sustainable Development talk
- 13:00-14:00. Lunch break
- 14:00-15:00. Group activity
- 15:00-16:00. Whole room discussion
- 16:00-16:30. Coffee break
- 16:30-17:00. Wrap up and future work

A. Invited Speakers' Profiles

To provide diverse perspectives on sustainability in HRI, we will invite four speakers for 15-minute flash talks. In line with our commitment to fostering local expertise and reducing travel-related carbon footprints, we will prioritize Australia-based speakers for these roles. The speaker profiles will cover the following key areas:

1. Equity and Access - Technology for All: the crucial intersection of technology and social equity. E.g.: collaborative projects with underserved communities or Indigenous groups to create culturally appropriate technological solutions; strategies for ensuring equitable access to technology in diverse socioeconomic contexts. Confirmed speaker: Linh Do, University of Melbourne.

2. Data and Computational Sustainability: the role of data and computational methods in advancing climate and sustainable solutions. E.g., leveraging data for climate-responsive technology; data-driven lifecycle assessment of technologies. Confirmed speaker: Monica Danilevicz, University of Western Australia

3. Ethics in Sustainable Technology: The moral and societal implications of sustainable technology development. E.g., ethical frameworks and policy considerations for responsible AI innovation; balancing technological advancement with environmental and social responsibility. Confirmed speaker: Fahimeh Abedi, University of Melbourne.

4. Sustainable Design and Critical Thinking: challenging conventional design paradigms. E.g., integrating lifecycle thinking into technology development processes; critical analysis of "techno-solutionism" in sustainability discourse.

Additionally, during the group activity, we will open the floor for HRI researchers aiming to share positive examples of research on sustainability in HRI, in 5-minute pitches.

B. Before the workshop

We will conduct a survey among HRI researchers before the workshop date to collect information on what sustainability – in the context of HRI – currently means for researchers.

We will advertise the workshop through our own HRI networks (e.g., via LinkedIn posts) and to technology + climate networks (e.g., Climate Robotics Network, Climate Salad. We will communicate with attendees what they can expect from the workshop, and what we will expect from them, in a timely manner. We will disclose the predicted outcomes (see below) in advance, so that people may decide the extent to which they want to contribute. This will be particularly relevant in terms of paper authorship and ethical usage of ideas and solutions that might emerge during the workshop activities.

C. During the workshop

The data collected in the pre-workshop survey will be presented to attendees in the morning of the workshop, as means of introducing the topic and open questions that will be discussed throughout the day. The introduction session will also briefly outline the purpose of the workshop and the expected outcomes.

After the flash talks and panel, participants will be divided into groups of 3-4 people. Each group will be asked to imagine a desirable future where HRI has contributed to achieving one, or more, aspects of sustainability-related goals. Then, they will be asked to list any potential steps needed to achieve this future. As an example, groups would list any key stakeholders and resources involved in their future scenario (people, energy, materials, social norms). Then, they would identify unsustainable practices (e.g., wasteful energy use, inequality, short robot lifespans) and brainstorm sustainable alternatives (e.g., energy-efficient robots, longer-lasting hardware, more inclusive design). Groups would create a vision of what their chosen future looks like (both in terms of robots and human society). Then, they work backward to identify what steps or milestones would need to occur to make this vision a reality. They can focus on technological innovations, policy changes, or societal shifts. This will help us identify key sustainability challenges and solutions, inspiring concrete steps that HRI should take now to achieve future sustainability. Then, all participants will get together to discuss commonalities in these identified necessary steps, and derive a "sustainability manifesto" (akin to e.g., [7]) for HRI. Our goal is to translate this manifesto into an actionable statement that researchers could use to check whether their research is sustainable, or contributes to sustainability. The procedure of the group activity is illustrated in Figure 1. We took inspiration from science fiction prototyping and back-casting design methods [3], [8] for this hands-on activity.

We believe that this workshop will have the most impact if the people involved in the group activities have also learned from our invited speakers in the morning. Therefore, when signing up, we will ask attendees to commit to attending the workshop in its entirety.

III. After the workshop

We expect two tangible artefacts as a result of this workshop. Firstly, we will propose guidelines for a "sustainability statement" that can be used in future publications, similar

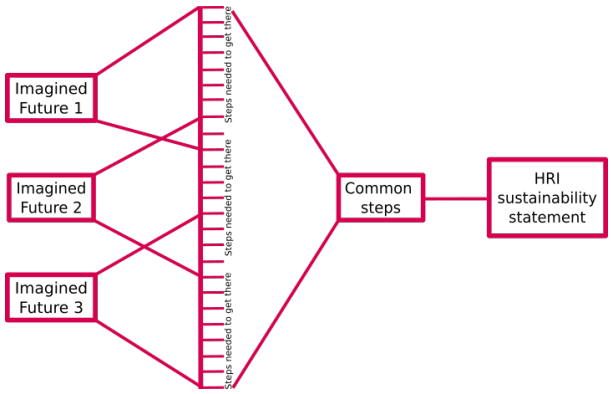


Fig. 1. Schematic of the afternoon group activity of backcasting imagined sustainable futures to understand needed steps to get there.

to how most publishers require statements regarding ethics or funding sources. Secondly, we will compile our findings regarding the current state of HRI research in relation to sustainability in a paper. This paper will both consist of work done in preparation for the workshop and the contents of the discussions during the workshop. Participants in the workshop will be invited to actively contribute to the completion of both these two artefacts.

In addition, we expect to see several outcomes that are less tangible while still valuable to the HRI community. The concept of sustainability can be intimidating and far from trivial to apply to HRI, so with this workshop we hope to reduce the barrier of entry to the topic while communally investigate what sustainability can mean in an HRI context. We hope that this workshop will constitute a platform for researchers in HRI with an interest in sustainability to meet, interact, and share experiences, ideas, and opinions. The activities in this workshop are intended to provide inspiration regarding how to think about, and work with, sustainability in an HRI context.

A. Ethics

One of the intended outcomes of this workshop is to document emerging reflections and ideas in a future paper. We acknowledge that topics of sensitive nature might be mentioned. We will have a dedicated note-keeper who will be instructed to not write down anything sensitive that might be disclosed during the discussions. We are also committed to dealing with these collected data in a transparent and ethical manner. Participants will be told beforehand that the content of the discussions will be recorded, and will be offered the possibility to contribute to the future paper.

IV. Organisers

- **Ilaria Torre** is an assistant professor in Human-Robot Interaction at Chalmers University of Technology. Her research broadly covers verbal and nonverbal communication for HRI, and she is interested in developing HRI-specific sustainable research practices. She has served as local co-chair for HRI 2023, and sustainability co-chair for HRI 2025.
- **Sarah Schömbs** is a PhD student at the University of Melbourne, researching human-centred risk communication in Human-Robot Interaction and works as a research assistant on integrating sustainability into the IT Masters program. She is a sustainability co-chair for HRI 2025 and a Wattle Fellow at the University of Melbourne.
- **Katie Winkle** is an assistant professor in Social Robotics at Uppsala University, Sweden. She works with the power of, and embedded in, Human-Robot Interaction design and practice, with a keen interest in figuring out how to realise high level ethics guidelines in real robotics practice.
- **Sara Ljungblad** is an associate professor in Interaction Design at University of Gothenburg and Chalmers University of Technology, Sweden. Her research focuses on critical robotics and the role of design knowledge in research, to critically assess alternative sustainable design solutions.
- **Erik Lagerstedt** is a postdoctoral researcher at the University of Gothenburg, Sweden. He is particularly interested in the mundane situations and aspects of interaction with robots, including consequences of robot design and deployment in terms of ethics and sustainability.
- **Maria Teresa Parreira** is a PhD student in HRI at Cornell University, advised by Profs. Wendy Ju and Malte Jung. Her work focuses on machine learning for social competence in robots. She is a European Climate Pact Ambassador and Youth Delegate for the Major Group of Children and Youth, a UN-mandated MG.
- **Hannah Pelikan** is a postdoctoral researcher at Linköping University, Sweden. Her research documents encounters between humans and robots in public urban settings and raises questions about social sustainability.

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